

Thermodynamic State Vector Discrepancy: Absolute Difference Math Function in Biogeochemical Simulation Engines

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Web of Life

Sprint 073

Abstract

In complex ecological and thermodynamic simulations, maintaining rigorous adherence to mass conservation laws and homeostatic steady states is paramount. Sprint 073 introduces the pure, isolated helper function `computeAbsoluteStockDelta(actual, expected)` within the Web of Life simulation framework (`src/thermodynamics/state_validator.ts`). This function computes the absolute elemental stock discrepancies between actual and expected thermodynamic state vectors across closed elemental budgets (C, N, P, H_2O). By formalizing state discrepancies mathematically, the system provides foundational validation support for error-correction monads and exergy dissipation tracking. Official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

1 Introduction & Thermodynamic Foundations

The Web of Life simulation architecture models ecosystems as open thermodynamic systems bounded by material conservation and energy dissipation laws. To ensure simulation fidelity, state vectors representing biogeochemical pools must be continuously validated against expected theoretical equilibria.

- **First Law (Matter Conservation):** State vector discrepancies must be tracked strictly across closed elemental budgets without spontaneous generation or destruction of mass.
- **Second Law (Entropy & Dissipation):** Discrepancies quantify deviation from steady-state homeostatic equilibrium, feeding directly into dissipation and error-correction monad structures.

2 Mathematical Formalization

Let an elemental stock state vector be represented as a mapping from an elemental key $e \in \mathcal{E}$ to its scalar mass or molar stock $S_e \in \mathbb{R}$.

Given an actual state vector $\mathbf{S}_{\text{actual}}$ and an expected state vector $\mathbf{S}_{\text{expected}}$, the absolute stock delta for any elemental key e is defined as:

$$\Delta_e = |S_{\text{actual},e} - S_{\text{expected},e}|$$

Where missing keys in either vector default to an absolute stock value of 0. Total mass imbalance across the system is subsequently derived via vector norm aggregation:

$$\|\Delta\|_1 = \sum_{e \in \mathcal{E}} \Delta_e$$

3 Implementation Specification

The helper function is implemented as a stateless, deterministic TypeScript module located in `src/thermodynamics/state_validator.ts`:

Listing 1: Implementation of `computeAbsoluteStockDelta`

```
import { StateVector, ElementalKey } from './types';

/**
 * Computes the absolute elemental stock discrepancies between actual
 * and expected thermodynamic state vectors.
 *
 * @param actual - The measured or simulated current StateVector.
 * @param expected - The baseline or theoretical target StateVector.
 * @returns A record mapping each ElementalKey to its absolute scalar
 * discrepancy.
 */
export function computeAbsoluteStockDelta(
  actual: StateVector,
  expected: StateVector
): Record<ElementalKey, number> {
  const result = {} as Record<ElementalKey, number>;

  const allKeys = new Set<ElementalKey>([
    ...(Object.keys(actual) as ElementalKey[]),
    ...(Object.keys(expected) as ElementalKey[])
  ]);

  for (const key of allKeys) {
    const actualVal = actual[key] ?? 0;
    const expectedVal = expected[key] ?? 0;
    result[key] = Math.abs(actualVal - expectedVal);
  }

  return result;
}
```

4 Verification & Testing Plan

Unit tests implemented in `tests/sprint_073.test.ts` validate:

1. **Exact state matches:** Confirming zero deltas when actual equals expected.
2. **Stock deviations:** Handling positive and negative variances to compute accurate absolute values.
3. **Sparse vectors:** Robust handling of partial or missing elemental keys under strict type safety.

5 Conclusion

Sprint 073 establishes a robust, mathematically rigorous validation primitive for thermodynamic state vectors within the Web of Life ecosystem. By integrating pure stock delta calculations into error-correction pipelines, the architecture ensures unyielding adherence to the First and Second Laws of Thermodynamics across all simulation epochs.